HIFI-X12
Instructions

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Operating instructions

Warning

This sign indicates an important warning that serious injury or death may result if not paid attention.

Please stop using immediately when there is a problem

Please send this product back to the designated sennuopu dealer or nearby sennuopu repair station for testing.

Do not use any features that will distract you from driving

Any machine function operation that affects attention must wait until the car is completely stopped. To use these functions, it is necessary to park the vehicle
in a safe area first, otherwise, accidents may occur

The volume must be kept at a level where the noise outside the vehicle can be heard while driving

If the volume is too loud to make the warning sounds of emergency vehicles and road warning signals (train intersections, etc.) inaudible, it is dangerous and may cause accidents; playing music at a high volume in the car may also cause hearing damage.

Do not decompose or transform

Otherwise, accidents, fire or electric shock may occur.

Do not let children touch small objects such as bolts or screws

Swallowing small objects can cause serious injury. If swallowed by mistake, please call a doctor immediately。

Please apply this product to 12V battery car

The application of other types of vehicles may cause fire, electric shock or other injuries.

Pay attention to

This sign indicates an important warning that injury or damage to the equipment may occur if you do not pay attention.

Safety

Product cleaning

Please use a dry soft cloth to clean the product regularly. If the dirt is difficult to remove, please wet the soft cloth with water only. Any other solvent may dissolve the product.

Temperature

Before starting the machine, make sure that the temperature in the car is between – 20℃ and + 60℃

Repair

If you encounter problems during use, please do not handle them by yourself. Please return this product to the designated sennuopu dealer or nearby sennuopu repair station for inspection or repair

Install

Make sure the product is not installed in any of the following places:

- Near direct sunlight and heat sources
- High humidity or near water source
- Dusty place
- Place of violent vibration

Pay attention to

Sennuopu shall not be responsible for the loss of data, even in the process of using this product。

Read

Preparation

The HIFI-X12 machine has a matching wire controller, which can be used to control the HIFI-X12 machine easily. Wechat small programs or computer software can also be used to adjust the host audio source switching, EQ and sound effect data storage or call, etc. As the control modes of the three modes are different, they are described in different chapters. The specific function list is as follows:

Feature List

Catalog		Position		
		Applets	PC	Controller
Basic operation	Control interface	4	8	13
	Machine connection	4	8	13
	Sound source selection	4	9	-
	Preset Sound Settings	4	10	-
	Volume adjusting	4	10	13
Channel operation	Replace the output channel	5	10	-
	Set Channel EQ	5	10	-
	Set equalizer	5	11	-
	Set channel volume and phase	6	11	-
	Set up frequency divider	6	11	-
	Configure Output Signal Function	6	12	-
Delay	Set delay	7	12	-

Note: music playing priority: Bluetooth, U Disk .

Data connection priority: computer software, mobile phone software .

Spare parts list

According to the model you purchased, the packing box should be equipped with the following accessories. If there is any shortage, please inform the dealer or manufacturer as soon as possible.

Parts	Quantity	HIFI-X12
USB cable	1	●
USB drive	1	●
Control line	1	●
Controller	1	●
Fixing bracket	2	●
High Level Line	1	●

USB cable

USB drive

Controller

Control line

Fixing bracket

High Level Line

Software Instructions

The mobile phone downloads the **HIFI-X12** mobile phone software installation by scanning the QR code on the manual.

After successfully installing **HIFI-X12** tuning software on the mobile phone, open the **HIFI-X12** machine and connect to Bluetooth. For the first time, open the mobile phone "Settings" → "turn on Bluetooth" → "search device" → " **HIFI-X12 - BLE**".

Then open the **HIFI-X12** mobile phone software, click " " to enter Bluetooth connection → click " **HIFI-X12 - BLE** " → connect the software to the machine.

1、General user interface

Home page interface

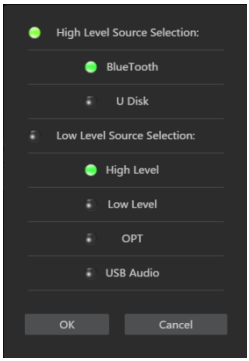
●"中" in the main interface is the language option, which can be in Chinese or English.

Volume adjusting

●Main volume size display, volume adjustment control

Sound source selection

●Click the sound source setting to select the type of input sound source.



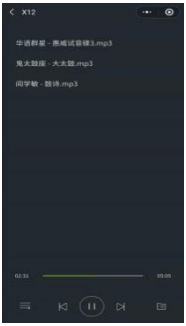
Note: due to the need to retain the warning sound of the reversing radar of the original car, the high-level input sound needs to be suspended no matter which input mode is selected, otherwise cross-talk mixing will occur.

Preset Sound Settings

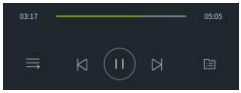
●There are 6 groups of sound effect data inside the machine, which can be directly set and saved by users.

U disk play

. When you use USB flash disk to play, you can view and control the playback status through " " in order to play normally, you need to select U disk in the audio source.

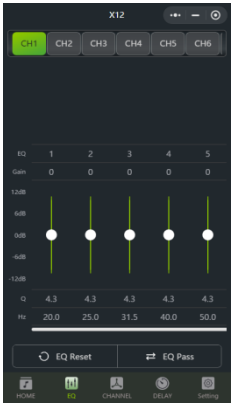


Users can freely select songs in the menu and control the menu playing sequence



2、Professional user interface

Click [**EQ**] to enter PEQ interface.

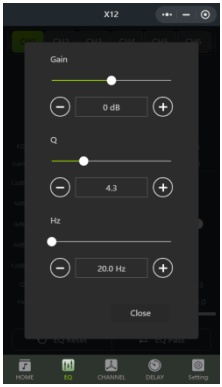


Select Output Channel

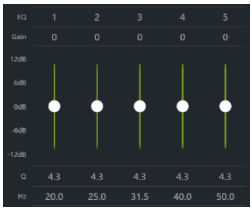
●Output channel selection: there are 10 output channels in total In this area, you can click to select different channel outputs.

Set Channel EQ

●Each channel has 31 sections of EQ adjustable. Click the frequency to be adjusted to enter the frequency gain and Q value for adjustment.

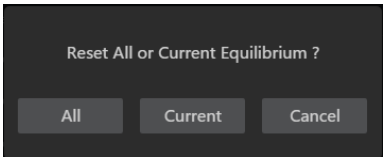


●The gain can also be adjusted by the following icon, where the gain can be adjusted from - 12dB to + 12dB.

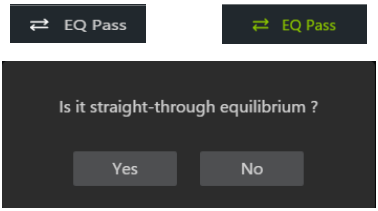


Set equalizer

- Click "**All**"in [**Reset**] to reset the parameters of all channels; "**Current**" to reset the parameters of currently selected channels.



•Click [**EQ Pass**], the EQ parameter adjusted by the current channel is invalid, but will not be cleared, which is used as the sound effect comparison before and after tuning.

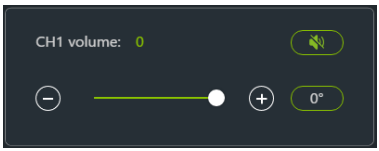


3、Click [Channel] to enter the interface

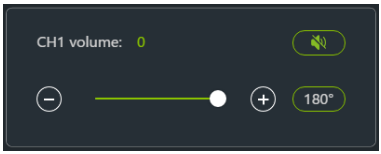
Set channel volume and phase

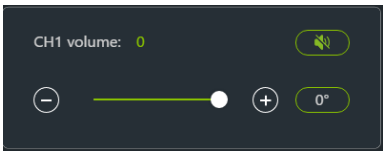
a. Channel output volume setting, click " " to set whether the output channel is mute, and display " " as mute status.

b. Drag slider to set output volume.



c. Click [**0°**] or [**180°**] to change the output phase of the output channel.





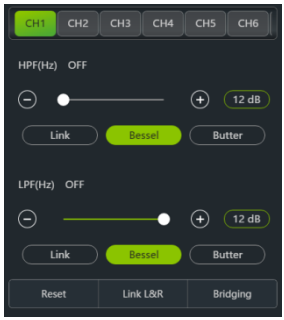
Set up frequency divider

a. Frequency divider: each channel is equipped with independent high pass and low pass filters.

b. The range of filtering frequency division point is 20Hz ~ 20kHz. You can click the corresponding value to set the frequency

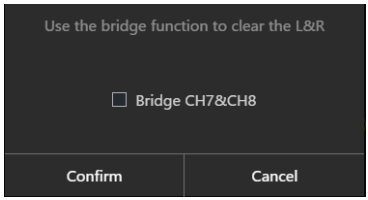
c. The selection range of filtering slope (slope) is: 6/12/18/24/30/36/42/48dB

d. There are three filtering algorithms of "Link ,Bessel and Butter"

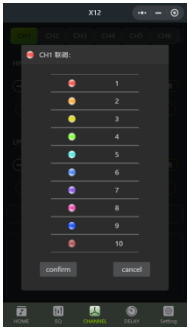


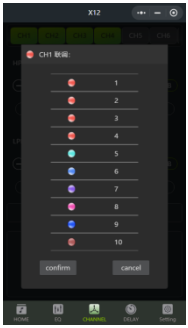
Configure Output Signal Function

•Click [**Bridging**] to bridge the ch7 and ch8 channels. The bridging method: connect the positive pole of audio cable to the positive pole of ch7 and the negative pole to the negative pole of ch8. Then, from the bridging option on the tuning software, check "**bridge ch7 & ch8**", and click OK. The sound effect adjustment after bridging is directly controlled by ch7.



•Click [**Link L&R**] to customize multi-channel simultaneous tuning. Multi channel joint debugging: any multiple channels can be combined and adjusted. Click the channel until the color of the required joint debugging channel is consistent, and then click confirm.

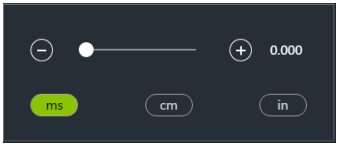




4、Delay Function Settings

a. Precise delay: adjust the delay of sound through addition and subtraction

b. The delay unit is [**ms**] by default, or [**cm**] or [**in**].



5、Setting interface

•Active noise gate Threshold: filter the detected audio noise (factory set, do not operate by yourself).

•Restore factory settings: restore the system to its original factory state.

PC Software Instructions

By downloading the attached loading data U disk to the computer and installing the **HIFI-X12** software on the computer, you can adjust the host sound source switching, EQ and sound data storage or call on the **HIFI-X2** model on the computer software.

Remarks: 1. The above mobile software can refer to the computer adjustment mode, and the function of the mobile software is the same as that of the computer software.

2. When the computer is connected to the tuning, it is not possible to use the mobile phone to connect the tuning at the same time. Whether the side will cause confusion and loss of parameters.

1、Precautions for software installation

- Software can be executed under Microsoft ® Windows ® system

The recommended configuration is as follows:

Operating system:Windows XP、Windows Vista、Windows 7、Windows 8、Windows 10。

CPU：1.6GHz Or higher.

Memory card: 1GB or higher。

Hard disk: 512MB or more free space

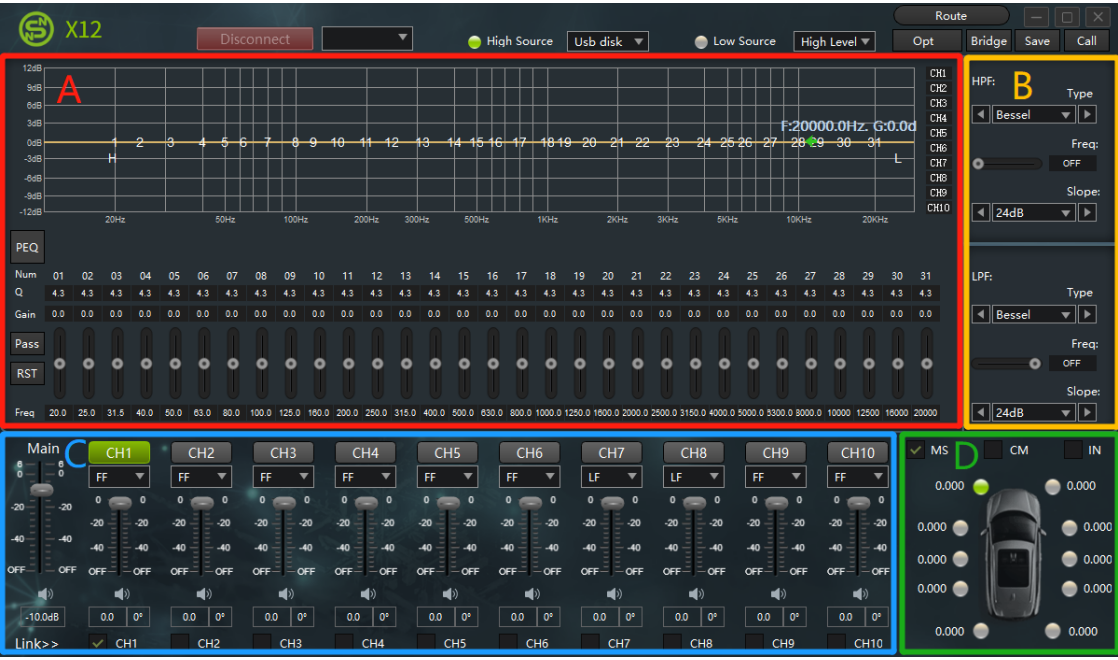
Computer resolution: 1280 * 768 or higher

- Before connecting the HIFI-X12 model to the computer, please install the HIFI-X12 computer software correctly.

2、PC software introduction

Control interface

The computer software interface can be divided into four areas: A: equalizer area, B: filter area, C: volume control area, D: delay control area

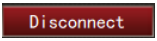


3、PC software operation instructions

Function setting

Machine connection

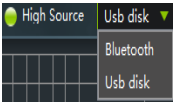
a. Before online, the interface displays "**Disconnect**"； after online successfully, the interface displays "**connected**".

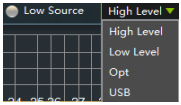


Sound source selection

Click to select the input source type.

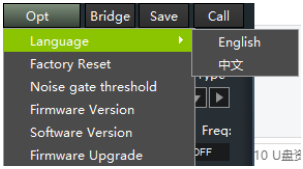
There are six kinds of sound source types to choose from. Among them, Bluetooth and USB flash disk are of high priority. When the machine detects that Bluetooth or USB flash disk is connected to play music, it will be forced to switch from other audio sources to Bluetooth or USB flash disk. After Bluetooth or USB flash disk stops playing, it will automatically switch back to the previous audio source. (Note: since the warning sound of the reversing radar of the original car needs to be retained, the high-level input sound needs to be suspended no matter which input mode is selected, otherwise cross-talk mixing will occur.).



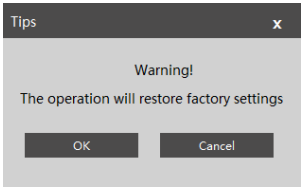


Function options

a. [**Opt**] click [**Language**] to display the status of English status text .



b. [**Factory Reset**] can restore all data to factory settings



b. [**Noise gate threshold**] filter out the detected audio noise (please do not operate by yourself since it has been set at the factory).

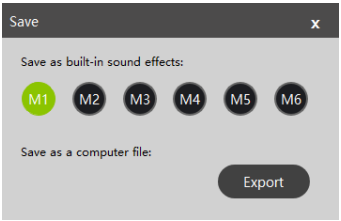
In the process of using **HIFI-X12**, the user can use this function to obtain the threshold value of noise door again in case of audio interruption played by a speaker. Method of use: turn the volume of the host to the minimum, click "**get**", and when a new threshold value appears, click "**set**".

c. [**Firmware Version**] is used to view the version number of the current machine firmware, and to determine whether the firmware version number needs to be updated.

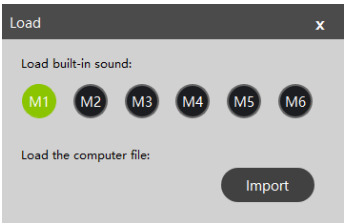
d. [**Software Version**] is used to view the version number of the current machine software and determine whether the software version number needs to be updated.

Preset Sound Settings

Click **Save** to save the adjusted sound parameters. The system provides 6 built-in modes for customers to save. (you can also save the sound effect to the computer through [**Export**])

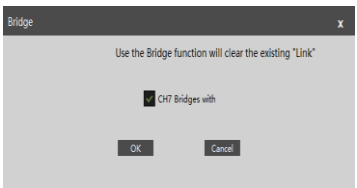


Click [**Load**] to select the saved sound parameters, or select the sound files saved to the computer through [**Import**].



Channel bridging settings

Click [**Bridge**] to bridge the ch7 and ch8 channels. The bridging method: connect the positive pole of audio cable to the positive pole of ch7 and the negative pole to the negative pole of ch8. Then, from the bridging option on the tuning software, check "bridge between ch7 and ch8", and click OK. The sound effect adjustment after bridging is directly controlled by ch7.

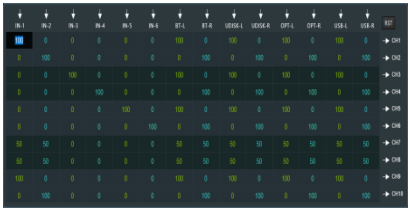


Channel routing settings

[**Route**] Can make any kind of sound source input can set any proportion at will, output from any one or more channels of 10 channels, achieve the purpose of mixing by adjusting the volume of each sound source in the channel, and set the mixing proportion range from 0 to 100.

Example 1: when the host audio signal input is only 4 channels but less than 6 channels, there is no sound in channels 5 and 6. At this time, the input position of channels 5 and 6 can be changed to 1 and 2, so that channels 5 and 6 will also have signal output without input;

Example 2: when it is necessary to connect a central speaker on the instrument panel, this function can make the wiring do not need to connect the left and right audio lines. After changing the mixing ratio of any two input channels to 50:50, the modified channel can be connected.



Main volume adjustment

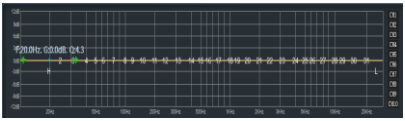
- Editing, adjusting and displaying area of total volume of output signal.

a. Adjust the **main volume**: directly drag the total volume push with the mouse to adjust the total volume, or directly input the value, scroll the mouse wheel and other ways to adjust the total volume, the adjustment range: + 6 ~ - 60.

b. Mute button: click the [ ] button in the main volume setting area to mute the total volume, and click again to cancel the mute (the mute will be automatically cancelled when the total volume is adjusted).

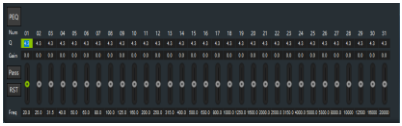
Set Channel EQ

1. Output signal channel frequency response curve display and frequency division balanced drag editing area.



There are 31 EQ sections that can be adjusted. When the mouse moves to the SN bit, press and hold the left key. Drag up and down to adjust the equalizer gain. Drag left and right to adjust the equalizer frequency. When the mouse moves to the left and right green arrows, press and hold Drag the left mouse button left and right to adjust the Q value of the equalizer; all adjustments will display the parameters of the current equalizer in real time under the serial number, and update the values in the parameter editing area of the equalizer.

2. Parameter display and edit area of output signal channel equalizer.



a. **Freq**: in the parametric equalization mode, in the editing area, the frequency can be adjusted by directly inputting the value, scrolling the mouse wheel and the up and down keys on the keyboard (adjustment range: 20Hz ~ 20kHz). In the graphic equalization mode, the frequency is the default 1 / 4 octave frequency distribution .

b. **Q value**: in the parametric equalization mode, in the editing area, you can directly enter the value, scroll the mouse wheel and the up and down keys on the keyboard to adjust the Q value (adjustment range: 0.4 ~ 20); in the graphic equalization mode, the Q value is the default 4.3.

c. **Gain**: in the editing area, it can be adjusted by directly inputting the value, scrolling the mouse wheel and the up and down keys on the keyboard (range: - 12dB ~ + 12dB)。

Set equalizer

- Equalizer mode

a. Click [**RST**] to return the equalizer parameters of the current channel or all channels to the initial state: uniform frequency distribution, Q value of 4.3, gain of 0dB.

b. Pass through Equalization: click [**Pass**] to turn all equalizers of the current channel into a non start state (**EQ PASS**)； click again to turn them into a start state.

Note: this operation will only change the gain of the equalizer.

Set up frequency divide

- Output signal filter editing area

Frequency setting: the low-pass or high-pass frequency can be adjusted by directly inputting the value, scrolling the mouse wheel, up and down keys on the keyboard, and directly dragging the LPF / HPF on the curve display area. The adjustment range: 20Hz ~ 20KHz.

Low/High pass filter slope：6/12/18/24/30/36/42/48dB

High / low pass has three filtering algorithms: Link, Bessel and Butter.

Set channel volume and phase

• Output channel volume, phase editing adjustment and display area.

Normal phase **Antiphase**

a. Channel volume: you can adjust the channel volume directly by dragging the push button with the mouse, or you can input the value. The volume can be adjusted by the way of the mouse wheel, and the adjustment range is 0 ~ - 60.

b. Mute button: click [ ] to mute the channel volume, and click again to cancel the mute (the mute will be automatically cancelled when the volume is adjusted) .

c. Click [**0°**] or [**180°**] to change the output phase of the output channel.

d. Each channel can be automatically frequency cut by the system, and the speaker can be easily divided into two or three frequencies. (HF：High frequency；MF：Mid frequency；LF：Low frequency；MHF：Medium treble frequency；MLF：Medium bass frequency；FF：Full Frequency)

Configure Output Signal Function

•By clicking and checking any multi-channel connection, 10-channel connection can be achieved.

Set delay

• Output signal delay regulation area

a. Precise delay: you can directly enter the value or scroll the mouse wheel and the up and down keys on the keyboard to adjust the delay time .

b. Delay unit: ms, cm, in.

Wire control instructions

●HIFI-X12 needs to connect the wired controller with the host before using the wired controller. When using the Udisk as the input source, the wired controller can control the selection of song menu, volume, cut songs and so on in the Udisk. Other sources can only control the volume.

Machine Connection

●Connect the HIFI-X12 wired controller to the machine for power supply. The wired controller connection interface has a separate dedicated interface on the host.

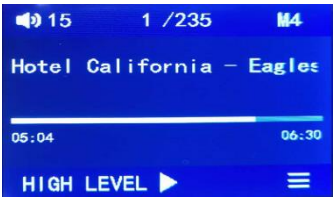
Control interface

●The in-line controller of the HIFI-X12 machine has the following control interface.

1.Master Volume Adjustment

a.By default, the rotate button adjusts the main volume (main volume 0-35), increases the volume clockwise, and decreases the volume counterclockwise..

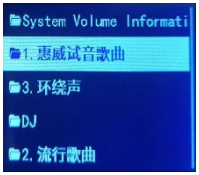
b. Press the volume adjustment button to pause/play the song..



C.【 】或【 】Or [Operate on the previous or next song of a song]

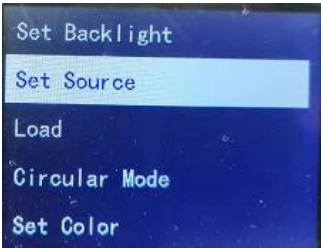
2. Song selection

●Press [SRC] to enter the main menu of Udisk, the volume knob is used as a selection column, press into the column, browse through the song with the volume knob, and press the volume knob again for the song selection.



2. set an option

●Press [ ] to enter the main menu of Udisk, the volume knob is used as a selection column, press into the column, browse through the song with the volume knob, and press the volume knob again for the song selection.



1、Interface introduction

1. Wire controller.

2. Square Port USB interface.

USB 2.0 cable can be connected to the computer for detailed tuning operation and setting

3. Optical fiber signal input.

Connect the car CD or DVD optical cable, switch the input sound source of DSP host to "optical fiber", and play the optical fiber signal

4. Car CD / DVD multimedia host.

5. Start mode switch.

When the switch is turned to the "ACC" end, the machine is started by ACC; when the switch is turned to the "high" end, the machine is started by the high-level input signal H1 - / H1 +

6. High level audio signal output interface.

7. 2 * 30A fuse.

8. U disk interface.

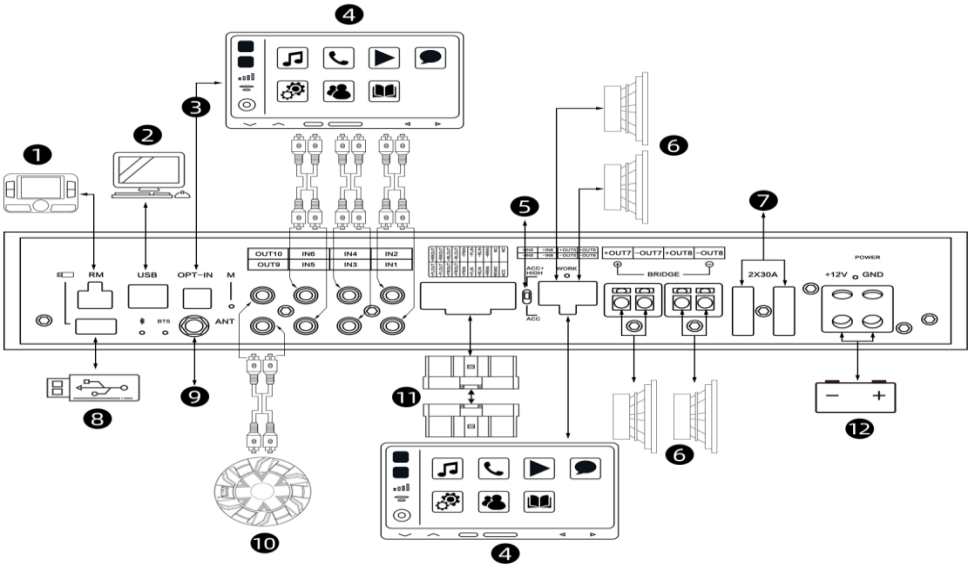
9. Bluetooth signal receiving antenna.

10. External active subwoofer.

11. 20pin universal DSP interface.

12. 12V battery.

2、Connection diagram



DSP T8 BASSPRO Technical parameter index

1、Technical specifications

RMS@4Ω	75W*6(AB class)+125W*2(D class)4Ω，THD＜1%；250W（D class，Bridge）4Ω，THD＜1%
MAX@4Ω	100W*6(AB class)+190W*2(D class)4Ω，THD＜1%；270W（D class,Bridge）4Ω，THD＜1%
Signal to noise ratio	≥101dB
Frequency gain	-12dB~+12dB
Frequency response	20Hz~20KHz
Input impedance	Low level input: 15KΩ；High level input:10 Ω
Phase position	0°/180°
Start mode	DC High level /ACC Input start
Working temperature	-30~70°C
Storage temperature	-40~85°C
Start high level input	High level start （H1-/H1+）
Activate ACC input	+12V Starting voltage
Power requirements	DC 14.4V (9V~16V) Negative ground
Standby current	750mA
Net weight	3.7kg(KG)
Volume	310mm (Width) *180mm (Depth) *57mm (Height)

2、Functional parameters

Input signal type	6-channel high level, 6-channel low level, 5.0 APTX-HD Bluetooth, USB sound card,U disk,OPT
Output Location	6-channel AB amplifier output + 2-channel class D amplifier output + 2- channel RCA output
Signal gain	Main volume + 6 ~ - 60
Filter Slope	CH1~CH10 6/12/18/24/30/36/42/48 dB Oct
Q value	0.4~20
Delay	0.000~20ms/0.000~680cm/0.000~268in
Scene saving	The machine can save 6 preset music scenes

Components of harmful substances

Name and content of toxic and harmful substances or elements in products

Part Name	Hazardous substances or elements					
	Lead (Pb)	Mercury (Hg)	Cadmium (Cd)	Hexavalent Chromium (Cr(VI))	Polybrominated Biphenyls (PBB)	Polybrominated Diphenyl Ethers (PBDE)
Circuit Board Component	×	○	○	○	○	○
Shell Component	○	○	○	○	○	○
Package Components	×	○	○	○	○	○
Attachments	×	○	○	○	○	○

This form is compiled in accordance with SJ/T 11364.

○ :Indicates that the content of this harmful substance in all materials of this part is within the limits set by GB/T 26572.

× :Indicates that the content of hazardous substances in at least one homogeneous material of the part exceeds the limit set by GB/T 26572

The content of "x" parts exceeds because there is no mature replaceable technology in the industry at present